

SUBMISSION C SOLUTION TO PROBLEM 10

ABSTRACT. We prove that if Γ is a finite join-irreducible simplicial graph with at least three vertices and if each tracial vertex algebra (M_v, τ_v) contains a trace-zero unitary, then the tracial graph product von Neumann algebra is properly proximal.

1. PRELIMINARIES

We use the following standard convention. A finite graph $\Gamma = (\mathcal{V}, \mathcal{E})$ is called *irreducible* if it is not a non-trivial join: equivalently, there is no non-empty proper subset $S \subset \mathcal{V}$ such that every vertex of S is adjacent to every vertex of $\mathcal{V} \setminus S$. Equivalently, the complement graph Γ^c is connected.

For $U \subset \mathcal{V}$, let Γ_U be the induced subgraph and let

$$(M_U, \tau_U) = \bigstar_{v \in \Gamma_U} (M_v, \tau_v)$$

denote the corresponding graph product von Neumann algebra. We put $M_\emptyset = \mathbb{C}$. The canonical trace-preserving conditional expectation is denoted

$$E_U : M \rightarrow M_U.$$

We write

$$\text{Lk}(v) = \{w \in \mathcal{V} : (v, w) \in \mathcal{E}\}, \quad \text{St}(v) = \text{Lk}(v) \cup \{v\}.$$

We use the definition of proper proximality for finite von Neumann algebras introduced by Ding–Kunnawalkam Elayavalli–Peterson [3]. If $N \subset M$ is a von Neumann subalgebra with expectation, let X_N denote the M -boundary piece associated with N , i.e. the boundary piece generated by the Jones projection $e_N : L^2(M) \rightarrow L^2(N)$. The small-at-infinity operator system is denoted $S_{X_N}(M)$. By [3, Theorems 6.2 and 8.5], M is properly proximal relative to X_N exactly when there is no M -central state on $S_{X_N}(M)$ whose restriction to M is normal.

We shall use two standard consequences of the theory in [3].

Lemma 1.1 (Upgrading from relative boundary pieces). *Let (M, τ) be separable and tracial. Let $X_1, \dots, X_n$ be M -boundary pieces and let q_i be the support projection of $K_{X_i}(M)$ in the bidual model of [3, Section 8]. Let $q_{\mathbb{K}}$ be the corresponding support projection for the compact boundary piece $\mathbb{K}(L^2 M)$. Suppose:*

- (i) M is properly proximal relative to every X_i ;
- (ii) $\bigwedge_{i=1}^n q_i = q_{\mathbb{K}}$.

Then M is properly proximal.

Proof. If M were not properly proximal, then [3, Theorem 8.5] would give an M -central state φ on the bidual small-at-infinity system for the compact boundary whose restriction to M is normal. Since proper proximality relative to any X_i rules out amenable direct summands, $\varphi(q_{\mathbb{K}}) = 0$; otherwise Connes' amenability criterion would produce an amenable direct summand.

For each i , if $\varphi(q_i^\perp) > 0$, then the normalized compression of φ by $q_i^\perp$ gives an M -central state on the small-at-infinity system relative to X_i , still normal on M , contradicting relative proper proximality. Hence $\varphi(q_i) = 1$ for all i . Since the q_i are central projections and the family is finite,

$$\varphi\left(\bigwedge_i q_i\right) = 1.$$

By assumption, $\bigwedge_i q_i = q_{\mathbb{K}}$, contradicting $\varphi(q_{\mathbb{K}}) = 0$. $\square$

Lemma 1.2 (Boundary cover for graph products). *For a graph product $M = \star_\Gamma(M_v, \tau_v)$ and subsets $U_1, \dots, U_n \subset \mathcal{V}$, let q_{U_i} be the support projection associated with the boundary piece $X_{M_{U_i}}$. Then*

$$q_{U_1} \wedge \cdots \wedge q_{U_n} = q_{U_1 \cap \cdots \cap U_n}.$$

In particular,

$$\bigwedge_{v \in \mathcal{V}} q_{\text{Lk}(v)} = q_\emptyset = q_{\mathbb{K}}.$$

Proof. The first assertion is the support-projection form of the graph-product bimodule calculation for subgraph algebras; see Charlesworth–de Santiago–Hayes–Jekel–Kunnawalkam Elayavalli–Nelson [2, Theorem 0.4, Theorem 5.4, and Proposition 5.5]. Equivalently, the M - M boundary piece generated by e_{M_U} intersects/fuses with the one generated by e_{M_W} exactly as the subgraph algebras intersect:

$$M_U \cap M_W = M_{U \cap W}, \quad E_U E_W = E_{U \cap W}.$$

Iterating gives the formula for finitely many U_i .

For $U_i = \text{Lk}(v_i)$ over all vertices, the intersection is empty because $v \notin \text{Lk}(v)$ for every v . Thus the meet is $q_\emptyset$. Since $M_\emptyset = \mathbb{C}$ and the boundary piece generated by $e_{\mathbb{C}}$ is exactly $\mathbb{K}(L^2 M)$, we have $q_\emptyset = q_{\mathbb{K}}$. $\square$

2. A RELATIVE PROPER PROXIMALITY CRITERION FOR AMALGAMATED FREE PRODUCTS

We need the following elementary relative version of the usual paradoxical-decomposition proof for free products.

Proposition 2.1. *Let*

$$(M, \tau) = (A, E_B^A) *_B (C, E_B^C)$$

be a tracial amalgamated free product over a common von Neumann subalgebra B . Suppose there are unitaries

$$u_1, u_2 \in C, \quad w \in A$$

such that

$$E_B^C(u_1) = E_B^C(u_2) = E_B^C(u_1^*u_2) = 0, \quad E_B^A(w) = 0.$$

Then M is properly proximal relative to the boundary piece X_B associated with B .

Proof. Let $A^\circ = \ker E_B^A$ and $C^\circ = \ker E_B^C$. In the standard L^2 -model of the amalgamated free product,

$$L^2(M) = L^2(B) \oplus H_A \oplus H_C,$$

where H_A is the closed span of reduced words beginning in A° , and H_C is the closed span of reduced words beginning in C° . Let P_A and P_C be the corresponding orthogonal projections.

A direct Fock-space computation shows that

$$P_A, P_C \in S_{X_B}(M).$$

Indeed, right multiplication by an algebraic reduced word changes the first letter only through cancellations that factor through $L^2(B)$; hence the commutators with the right action of M lie in the ideal generated by e_B , namely $K_{X_B}(M)$. This is the standard computation used in [3, Section 6].

Assume, toward contradiction, that there exists an M -central state φ on $S_{X_B}(M)$ whose restriction to M is normal. Since $u_1, u_2 \in C^\circ$ and $E_B(u_1^*u_2) = 0$, the subspaces $u_1^*H_A$ and $u_2^*H_A$ are orthogonal and both are contained in H_C . Hence

$$u_1^*P_A u_1 + u_2^*P_A u_2 \leq P_C.$$

By M -centrality and unitarity,

$$2\varphi(P_A) = \varphi(u_1^*P_A u_1) + \varphi(u_2^*P_A u_2) \leq \varphi(P_C).$$

Similarly, since $w \in A^\circ$,

$$w^*P_C w \leq P_A,$$

and therefore

$$\varphi(P_C) = \varphi(w^*P_C w) \leq \varphi(P_A).$$

Thus $\varphi(P_A) = \varphi(P_C) = 0$.

Finally, since $u_1 \in C^\circ$, the range of $u_1^*e_B u_1$ is contained in H_C , so

$$u_1^*e_B u_1 \leq P_C.$$

Again by centrality,

$$\varphi(e_B) = \varphi(u_1^*e_B u_1) \leq \varphi(P_C) = 0.$$

But $1 = P_A + P_C + e_B$, so $\varphi(1) = 0$, a contradiction. $\square$

3. GRAPH-PRODUCT INPUT

For every vertex v , the graph product admits the canonical amalgamated free product decomposition

$$M = M_{\text{St}(v)} *_{M_{\text{Lk}(v)}} M_{\mathcal{V} \setminus \{v\}}.$$

This follows directly from the graph-product normal form and is standard; see [1] and [2, Section 1.1].

We now verify that the hypotheses of Proposition 2.1 hold for each vertex.

Lemma 3.1. *Let Γ be finite, irreducible, and $|\mathcal{V}| \geq 3$. Fix $v \in \mathcal{V}$ and put $B = M_{\text{Lk}(v)}$. In the decomposition*

$$M = M_{\text{St}(v)} *_B M_{\mathcal{V} \setminus \{v\}},$$

there exist a unitary $w \in M_{\text{St}(v)}$ and unitaries $u_1, u_2 \in M_{\mathcal{V} \setminus \{v\}}$ such that

$$E_B(w) = 0, \quad E_B(u_1) = E_B(u_2) = E_B(u_1^* u_2) = 0.$$

Proof. Choose, for each vertex s , a trace-zero unitary $a_s \in M_s$.

First take

$$w = a_v \in M_v \subset M_{\text{St}(v)}.$$

Since $M_{\text{St}(v)} = M_v \overline{\otimes} M_{\text{Lk}(v)}$ over the commuting graph product decomposition, we have

$$E_B(w) = \tau_v(a_v)1 = 0.$$

Let

$$L = \text{Lk}(v), \quad W = \mathcal{V} \setminus (\{v\} \cup L).$$

The set W is non-empty. Indeed, if $W = \emptyset$, then v is adjacent to every vertex in $\mathcal{V} \setminus \{v\}$, giving the non-trivial join decomposition

$$\Gamma = \{v\} + \Gamma_{\mathcal{V} \setminus \{v\}},$$

contrary to irreducibility.

We distinguish two cases.

Case 1: $L = \emptyset$. Then $B = \mathbb{C}$. Since $|\mathcal{V}| \geq 3$, choose distinct vertices $r, s \in \mathcal{V} \setminus \{v\}$. Put

$$u_1 = a_r, \quad u_2 = a_s a_r.$$

Both are unitaries in $M_{\mathcal{V} \setminus \{v\}}$. Their scalar expectations are zero:

$$\tau(u_1) = 0, \quad \tau(u_2) = 0.$$

Moreover,

$$u_1^* u_2 = a_r^* a_s a_r.$$

If r and s are adjacent, this equals a_s , whose trace is zero. If they are not adjacent, it is a non-empty reduced word with trace-zero letters, hence has trace zero. Thus $E_B(u_1^* u_2) = 0$.

Case 2: $L \neq \emptyset$. We claim that there exist $\ell \in L$ and $r \in W$ such that ℓ is not adjacent to r . If not, then every vertex of L is adjacent to every

vertex of $\mathcal{V} \setminus L$ because vertices of L are adjacent to v by definition and, by assumption, to every vertex of W . Hence

$$\Gamma = \Gamma_L + \Gamma_{\mathcal{V} \setminus L},$$

again contradicting irreducibility.

Choose such $\ell \in L$ and $r \in W$. Put

$$u_1 = a_r, \quad u_2 = a_\ell a_r.$$

Since $a_\ell \in M_\ell \subset B$ and $r \notin L$, both u_1 and u_2 lie in $M_{\mathcal{V} \setminus \{v\}}$ and satisfy

$$E_B(u_1) = E_B(u_2) = 0.$$

Furthermore,

$$u_1^* u_2 = a_r^* a_\ell a_r.$$

Because r and ℓ are not adjacent, this is a reduced word involving the vertex $r \notin L$, so its expectation onto $B = M_L$ is zero. Hence $E_B(u_1^* u_2) = 0$. $\square$

Corollary 3.2. *For every $v \in \mathcal{V}$, the graph product M is properly proximal relative to the boundary piece $X_{M_{\text{Lk}(v)}}$.*

Proof. Apply Proposition 2.1 to

$$M = M_{\text{St}(v)} \underset{M_{\text{Lk}(v)}}{*} M_{\mathcal{V} \setminus \{v\}},$$

using the unitaries supplied by Lemma 3.1. $\square$

4. PROOF OF THE THEOREM

Theorem 4.1. *Let Γ be a finite simplicial graph with $|\mathcal{V}| \geq 3$. Let (M_v, τ_v) be a separable tracial von Neumann algebra for every $v \in \mathcal{V}$. Assume:*

- (a) Γ is irreducible, i.e. not a non-trivial join;
- (b) every M_v contains a trace-zero unitary.

Let

$$(M, \tau) = \bigstar_{v \in \Gamma} (M_v, \tau_v)$$

be the tracial graph product von Neumann algebra. Then M is properly proximal.

Proof. For every vertex v , Corollary 3.2 shows that M is properly proximal relative to the boundary piece $X_{M_{\text{Lk}(v)}}$.

Therefore M has no amenable direct summand: if pM were an amenable direct summand, Connes' amenability criterion would give an M -central state on $\mathbb{B}(L^2(pM))$, hence by extension an M -central state on every $S_{X_{M_{\text{Lk}(v)}}}(M)$ normal on M , contradicting relative proper proximality.

By Lemma 1.2,

$$\bigwedge_{v \in \mathcal{V}} q_{\text{Lk}(v)} = q_{\mathbb{K}}.$$

Thus the finite family of boundary pieces $\{X_{M_{\text{Lk}(v)}} : v \in \mathcal{V}\}$ satisfies the hypotheses of the upgrading Lemma 1.1. Consequently M is properly proximal. $\square$

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